Beyond Hearing:

Exploring the Impact of Hearing Loss on Depression and Quality of Life

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Advanced Methods for Data Analysis
Project Report

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1 Introduction

Hearing is a vital human sense that is essential for communication, social engagement, and overall life satisfaction. When hearing capability diminishes, people might face not just sensory restrictions but also wider psychosocial effects. While hearing issues are very common—especially in older individuals—they frequently garner less focus compared to other chronic health problems, despite possibly affecting mental health.

The **Beyond Hearing** project aims to investigate the link between hearing impairments and depressive symptoms, specifically looking at how this connection might change when considering various contextual elements. Instead of viewing hearing loss as a separate issue, this project considers it a health problem that can connect with demographic, socioeconomic, and behavioral traits.

This study utilizes data from the National Health and Nutrition Examination Survey (NHANES), a comprehensive, nationally representative dataset featuring self-reported hearing assessments, mental health metrics, and extensive demographic and lifestyle details. The depth of NHANES enables the investigation of the relationship between hearing status and depression across various modeling hypotheses.

This research' main hypothesis suggests that **hearing loss is linked to increased levels of depressive symptoms**. To assess this statement, a regression model will be developed and examined. The main model will initially investigate the direct link between hearing and depression. It will subsequently be broadened to include essential demographic, socioeconomic, and pharmaceutical usage factors, to assess if these elements obscure or influence the main relationship.

This project seeks to enhance comprehension of the strength of the link between hearing challenges and depression from various analytical viewpoints, while strengthening essential statistical and data analysis abilities through the use of actual health data.

2 Data Sources and Variables

2.1 NHANES Survey Description

For this project, we selected multiple NHANES datasets to perform a comprehensive statistical analysis. NHANES (National Health and Nutrition Examination Survey) is an extensive, nationally representative study aimed at evaluating the health and nutritional condition of adults and children in the United States.

These datasets form the basis of the project and are utilized throughout the analysis to investigate the connection between hearing problems and depressive symptoms, alongside the influence of pertinent demographic and behavioral factors. By integrating data from various NHANES questionnaires, we can create multiple regression models, depict trends in the data, and assess possible confounding variables to enhance our understanding of the relevant associations

2.2 MODEL 1: Demographic Data Set (DEMO_J.xpt)

This dataset includes fundamental demographic details about participants, such as age, gender, race/ethnicity, educational attainment, and household attributes. It is the dataset we used to analyze **Model 1** of our project since demographic factors are essential for segmenting analyzes and comprehending variations within populations. We transform categorical variables such as education level into descriptive labels to enhance interpretation and visualization. Here are the variables that stood out the most for us, and therefore possible candidates to use for Model 1:

- **Age (RIDAGEYR):** Reported in years, allowing stratification by life stage (e.g., children, adults, seniors). Age is often treated both as a continuous variable and in categories for group comparisons.
- **Sex (RIAGENDR):** Encoded as 1 for male and 2 for female. Sex is a key factor in most health-related analyzes.
- **Race/Ethnicity (RIDRETH1 / RIDRETH3):** Participants are classified into main racial/ethnic groups, such as Non-Hispanic White, Non-Hispanic Black, Mexican American and others. This variable is useful for assessing health disparities.

2.2.1 Age by Sex Distribution

Figure 1 is a horizontal bar chart that visualizes the age distribution of a population, segmented by sex into 10-year intervals. The data shows a similar distribution pattern for both males and females, with the most populous group being individuals aged 30-40. The counts gradually decrease for older age brackets (50-80), while the youngest cohorts (0-10 and 10-20) are the smallest.

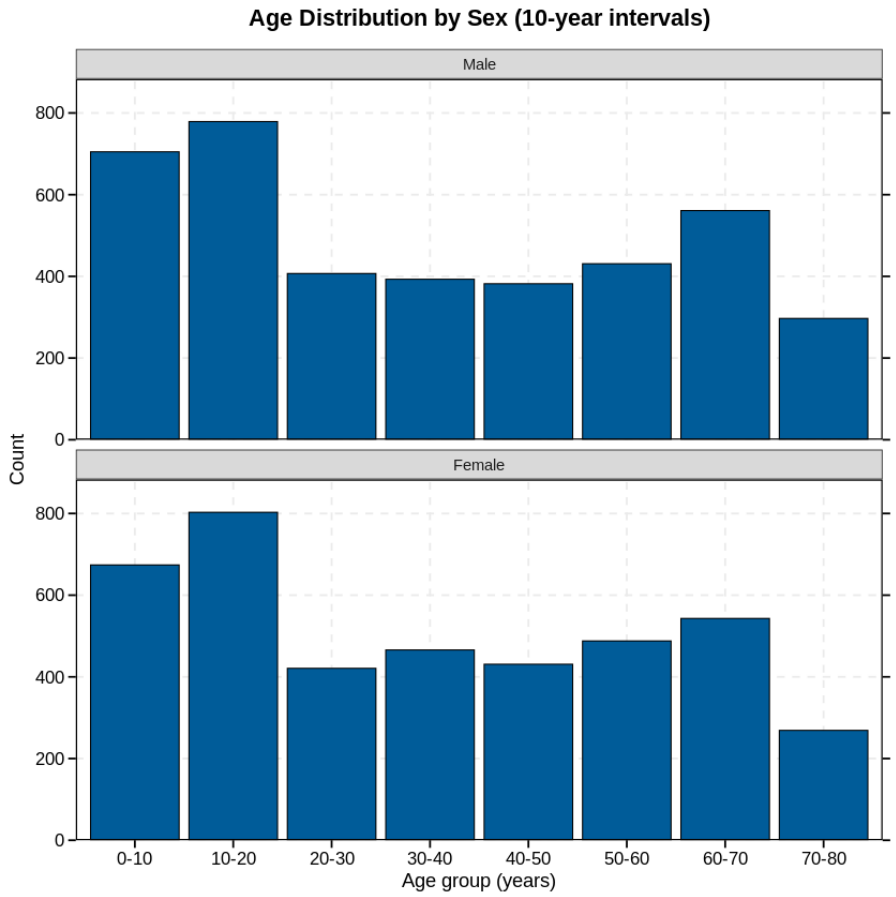


Figure 1: Age by Sex Distribution

2.2.2 Race / Ethnicity Distribution

Figure 2 is a bar chart that illustrates the distribution of participants based on ethnic and racial origin. "Non-Hispanic White" individuals form the largest group with 3,150 participants. The "Other Hispanic" group is the second largest with 1,367 participants, followed by the "Non-Hispanic Black" and "Mexican American" groups. The "Other / Multi-Racial" category is the smallest represented group, comprising 820 participants.

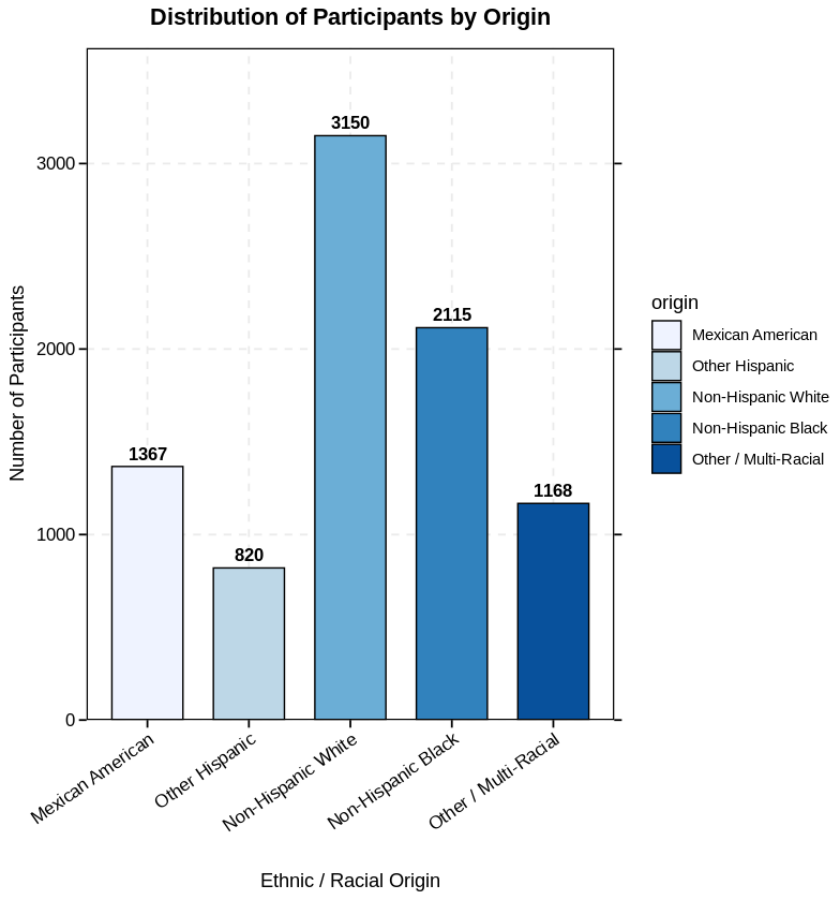


Figure 2: Origin Distribution

For Model 1, we finally chose to include age as the only covariate in the multiple regression analysis. This decision was motivated by the strong and well-established relationship between age and hearing ability, as well as age-related differences in depressive symptoms, after doing some research. We believe that age therefore represents a key confounding variable in the association between hearing difficulties and depression.

Although other demographic variables such as sex or race/ethnicity may also influence mental health outcomes, they were not included in this initial model in order to maintain model simplicity and focus on isolating the primary effect of hearing status. By controlling only for age, we believe Model 1 will provide a clear baseline assessment of whether the association between hearing difficulties and depressive symptoms persists after accounting for the most relevant demographic factor.

2.3 MODEL 2: Socioeconomic Controls (DEMO_J.xpt)

We incorporate socioeconomic variables in our analysis as control factors since they frequently affect health outcomes and behaviors. The demographic dataset used previously also contains variables that offer insights into the economic and social context of participants. The main socioeconomic factors that we might take into account are:

- **Level of Education (DMDEDUC2):** Indicates the upmost level or degree of education achieved. Categories were reclassified into descriptive terms like "Less than 9th grade," "9–11th grade," "High school graduate/GED," "Some college/AA degree," and "College graduate or above." Education serves as a substitute for knowledge, health understanding, and social standing.
- **Household Income (INDHHINC / INDFMINC):** Represents overall family earnings or income in relation to the poverty line. This variable aids in considering economic resources that could influence lifestyle decisions, healthcare access, and health results.

2.3.1 Level of Studies Distribution

Figure 3 shows the level of studies distribution. The largest group, at 32%, consists of individuals with a high school diploma or GED. This is followed by two groups each at 24%: those with a college degree or higher, and those with some college or an Associate's degree.

The smallest segments are those who completed 9th to 11th grade (11%) and those who completed less than 9th grade (9%).

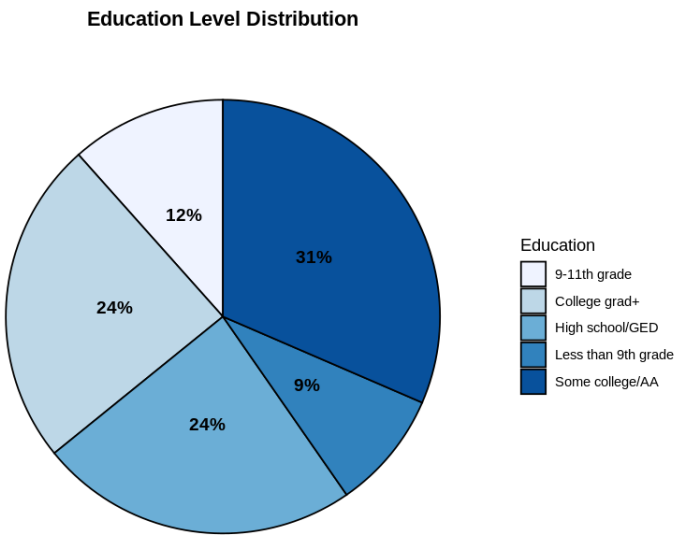


Figure 3: Studies Distribution

2.3.2 Income Distribution

Figure 4 is a pie chart that details the distribution of household income across eight distinct brackets. The most common income range is $25,000–$34,999, constituting 17% of the households. The highest income bracket ($75,000+) is the next largest segment at 16%. The lowest income bracket (¡$10,000) and the $35,000–$44,999 group are the smallest, each representing 9% of the total population.

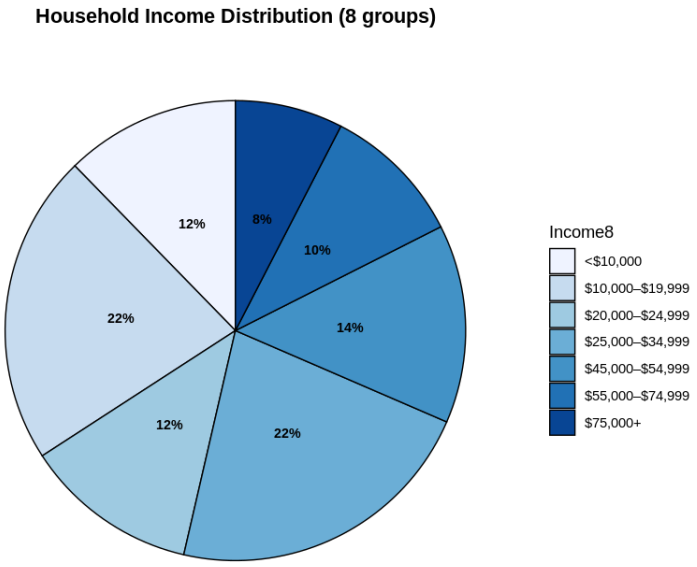


Figure 4: Income Distribution

For Model 2, we ended up focusing on annual household income as the main socioeconomic covariate in the multiple regression analysis. Household income was selected because it provides a direct measure of economic resources, which are closely linked to access to healthcare, hearing-related support, and mental health services, as well as overall living conditions that may influence depressive symptoms.

Although educational attainment is another important indicator of socioeconomic status, we believe it is often strongly correlated with income. Including both variables simultaneously could introduce multicollinearity and complicate the interpretation of the model coefficients. To preserve model parsimony and ensure clearer interpretation, we therefore chose to include household income alone as the socioeconomic control variable in Model 2. This approach allows us to assess whether the association between hearing difficulties and depression remains after accounting for economic status, while avoiding redundancy among closely related predictors.

2.4 MODEL 3: Smoking Habits & Drug Use (SMQ_J.xpt)

This dataset provides information on cigarette use among participants aged 12 years and older. It was used to analyze **Model 3** of our project, as understanding the relationship between smoking habits and drug use helps to characterize behavioral patterns within populations.

The primary variables in consideration for Model 3 are:

- **SMQ040 – Do you now smoke cigarettes?**: Indicates whether a participant currently smokes cigarettes every day, on some days, or not at all.
- **SMD030 - Age started smoking cigarettes regularly**: Indicates how old were the participants when they first started to smoke cigarettes fairly regularly.

2.4.1 Current Smoking Status Among Adults

Figure 5 indicates that more than half of adults (56.7%) are non-smokers, but a significant portion—over one-third (34.1%)—smoke daily. An additional 9.2% smoke occasionally, suggesting that nearly 43 of adults still engage in smoking to some extent.

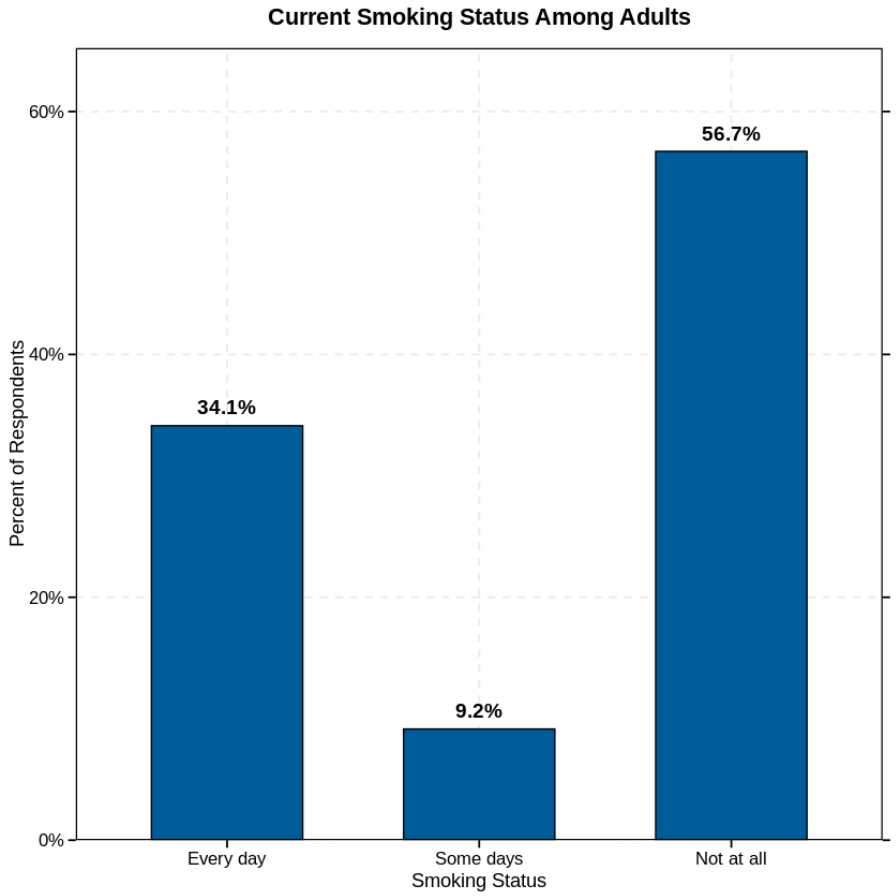


Figure 5: Current Smoking Status

2.4.2 Age at which participants started smoking regularly

Figure 6 selects age at smoking initiation as the smoking-related variable in the multiple regression analysis. This variable captures early-life exposure to smoking behavior, which may reflect long-term lifestyle patterns and cumulative health risks that are relevant for both mental health outcomes and hearing status.

In contrast, current smoking status among adults is highly heterogeneous, as it does not account for smoking duration, intensity, or past smoking history, as seen in the previous graph. Additionally, current smoking may be influenced by recent health conditions or behavioral changes, which could introduce reverse causality. To maintain conceptual clarity and avoid these limitations, we chose age at smoking initiation as a more stable and informative indicator of smoking-related exposure. This choice allows Model 3 to examine whether early smoking behavior contributes to the relationship between hearing difficulties and depressive symptoms.

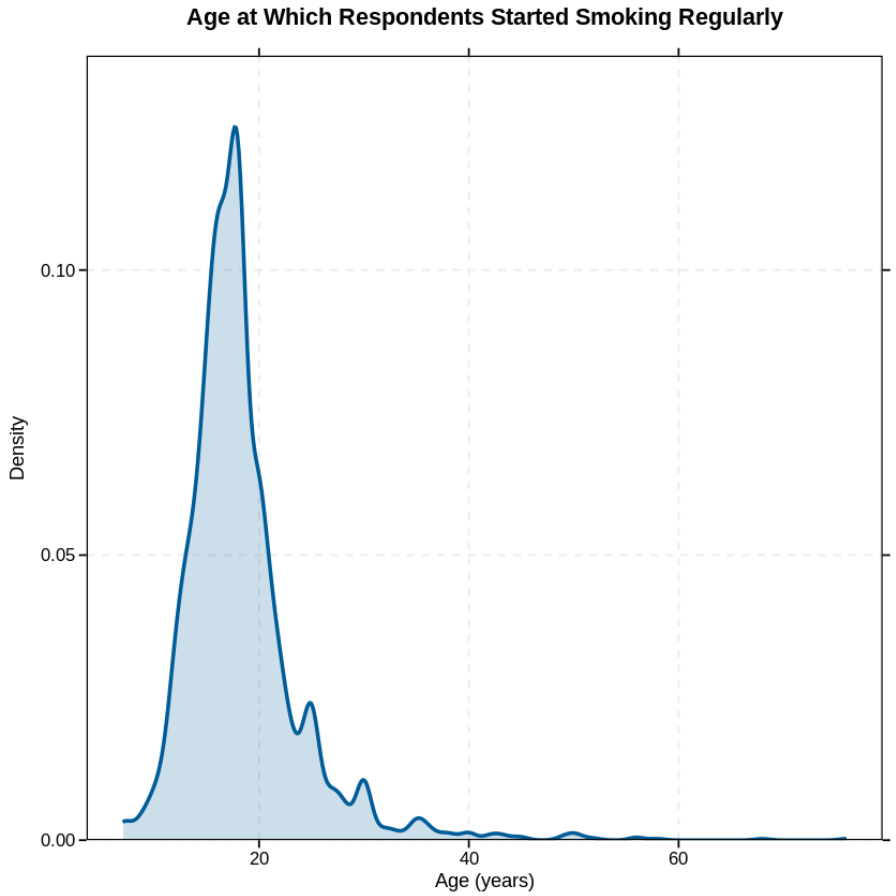


Figure 6: Current Smoking Status

2.5 Further Analysis of the Datasets

Prior to establishing the final modeling strategy, we performed an exploratory analysis of the chosen NHANES datasets. This part showcases a collection of descriptive and visual analyses that played a vital role in directing the project's focus and shaping important methodological choices. These graphs facilitated the recognition of significant patterns, evaluation of variability, and identification of potential confounding factors by visualizing distributions, bivariate relationships, and possible trends among the key variables of interest.

Though not all of these analyses are directly incorporated into the final regression models, they significantly influenced the study's direction and provided justification for the modeling decisions outlined in later sections.

2.5.1 Relationship Between Hearing Loss and Trouble Concentrating

Figure 7 displays the percentage of participants reporting difficulty concentrating by hearing condition. A clear increasing trend can be observed from participants with excellent hearing (10.7%) to those with moderate hearing trouble (29.6%), suggesting that difficulty concentrating becomes more prevalent with worse hearing conditions.

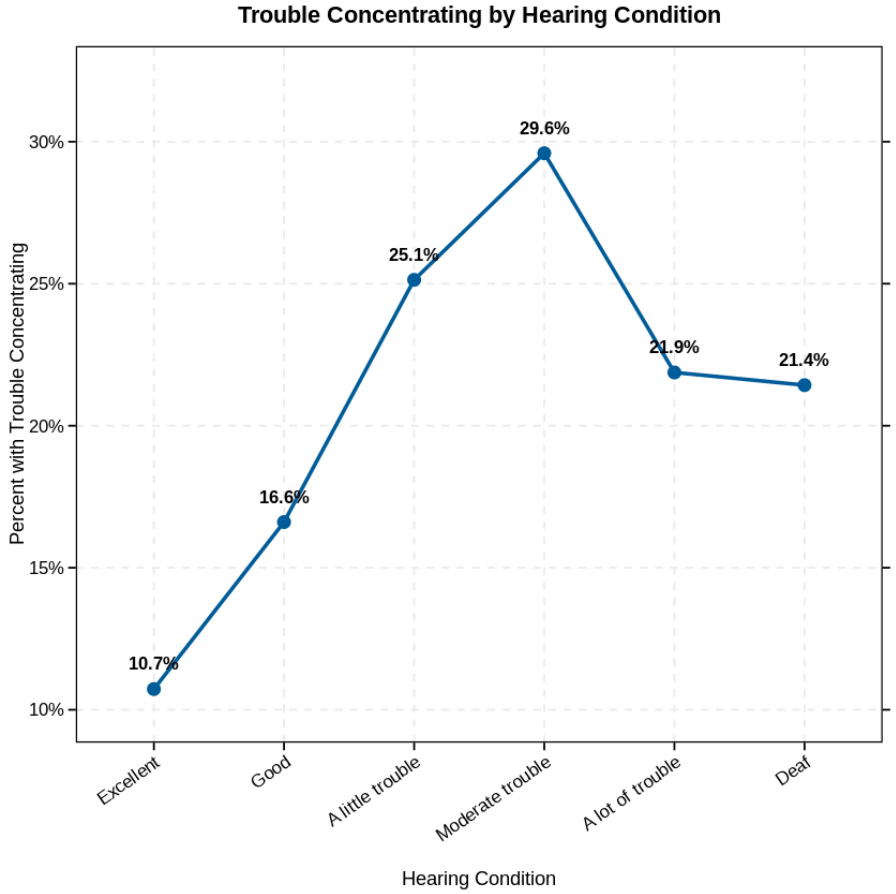


Figure 7: Trouble Concentrating by Hearing Condition

Figure 8 is a bar chart that shows a positive trend, indicating that participants reporting worse hearing conditions tend to exhibit higher average depression scores.

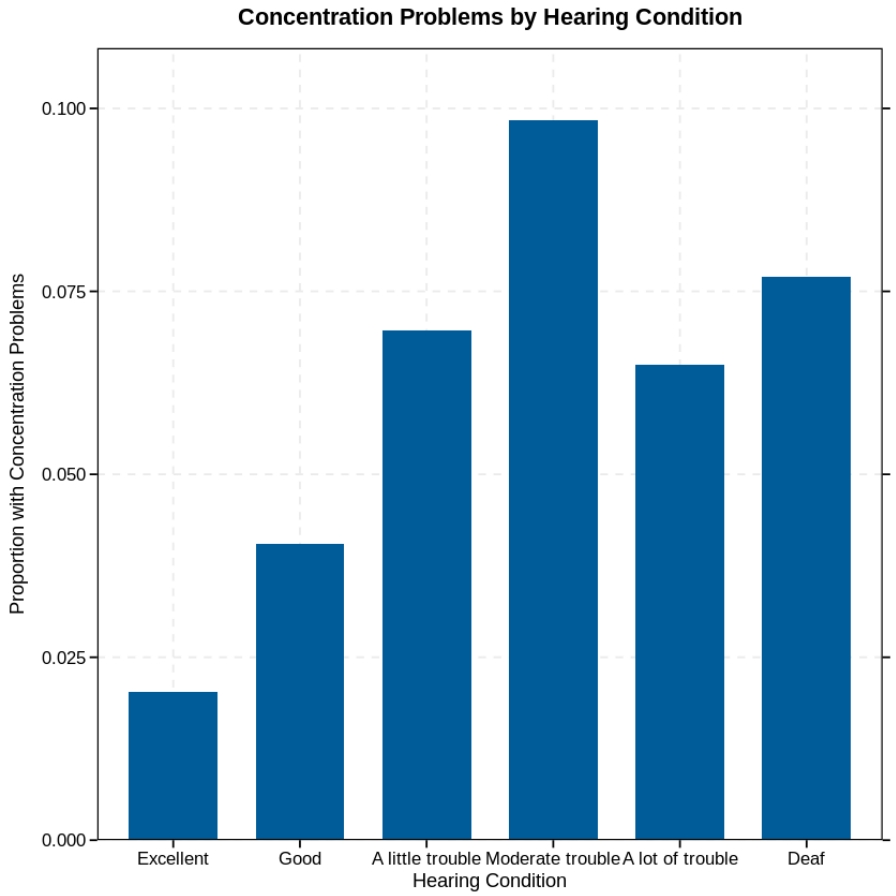


Figure 8: Concentration Problems by Hearing Condition Averages

2.5.2 Relationship Between Hearing Loss and Depression

Figure 9 shows a positive trend, indicating that participants reporting worse hearing conditions tend to exhibit higher average depression scores.

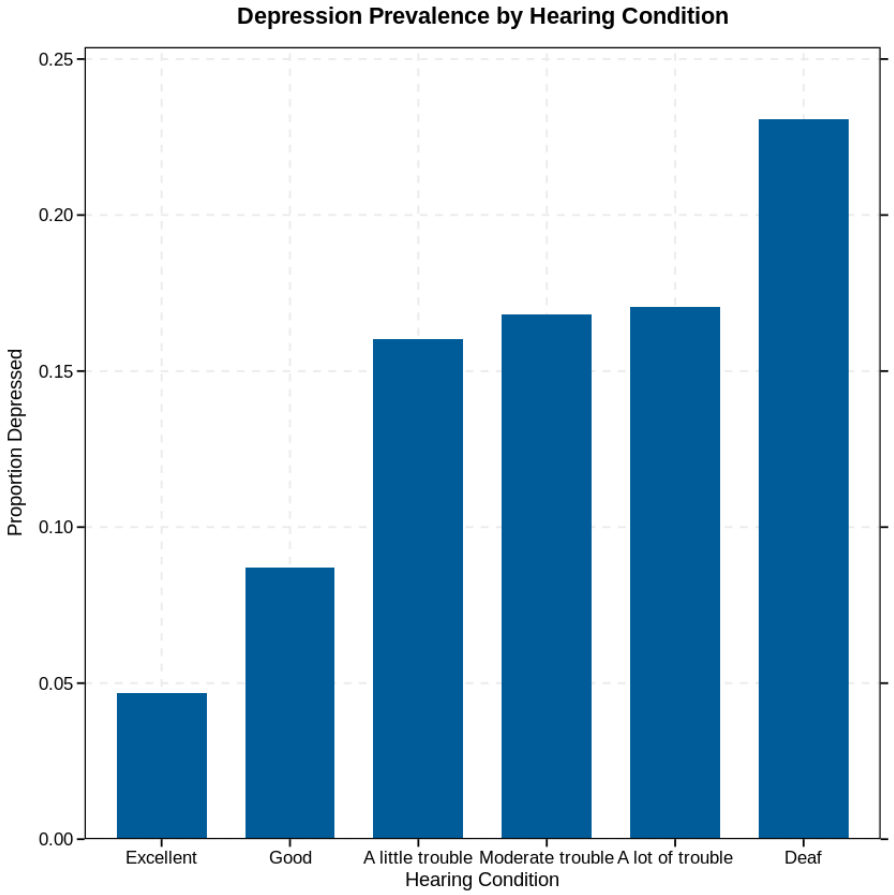


Figure 9: Average depression score by hearing condition

2.5.3 Depression by Hearing Status Across Age Groups

Worse hearing status may be associated with higher levels of depression. However, it is essential to consider age-related differences, as older adults are more likely to experience hearing difficulties, which does not necessarily imply the presence of depression.

To further examine whether age moderates the relationship between hearing loss and depression, depression scores were analyzed across different hearing conditions within distinct age groups.

Figure 10 indicates that among younger individuals (approximately 18-40 years old), higher average levels of hearing impairment are associated with higher levels of depression. In contrast, among older individuals, depression levels remain relatively lower despite worse hearing conditions.

These findings suggest that hearing difficulties are more strongly associated with depression in younger adults, whereas this relationship appears to be weaker or absent in older populations.

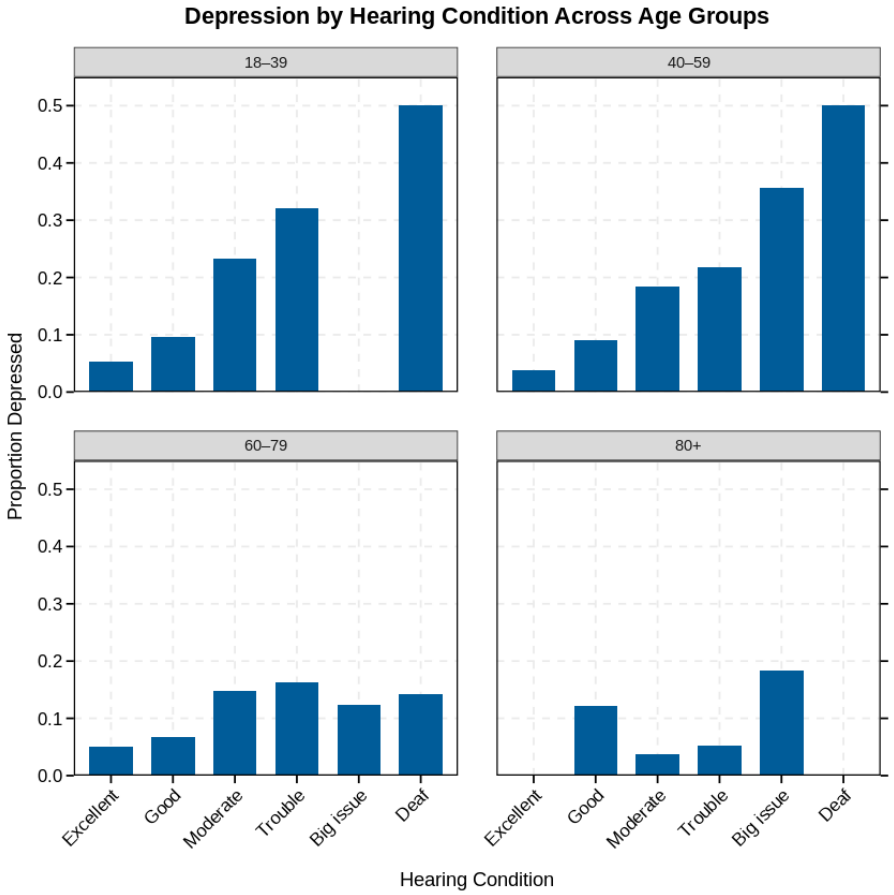


Figure 10: Depression by Hearing Condition Across Age Groups

3 Methodology

Prior to conducting further analyses, demographic information from the NHANES dataset was examined to characterize the study population and ensure the validity of subsequent data integration.

Participants were uniquely identified using the **SEQN** variable, allowing accurate rinformation across datasets.

3.1 Hearing Condition and Depression

The aim of this analysis was to determine if problems with hearing are linked to levels of depression and to evaluate if hearing can act as a significant predictor of depression. The analytical strategy adopted a systematic method, starting with an evaluation of correlation and advancing towards predictive modeling. This method enables the detection of statistically significant associations and assesses their practical importance for the research question, and enables us to do further research on the Models presented previously.

To start, a correlation analysis was performed to assess if there is a monotonic relationship between hearing and depression. Correlation was selected to account for the discrete characteristics of the hearing variable and the existence of duplicate values. The analysis of correlation produced a positive coefficient ($\rho = 0.138$), with a p-value less than 2.2×10^{-16}. This finding shows a statistically significant link between hearing and depression, suggesting that those who report more hearing issues are likely to have elevated depression scores. Nonetheless, the low value of the coefficient indicates a weak relationship, suggesting that hearing by itself does not significantly account for differences in depression levels. For the project, this outcome indicates that the relationship is present but weak, warranting additional modeling instead of concluding solely from correlation.

After identifying a relationship, a basic linear regression model was applied with depression score as the dependent variable and hearing as the independent variable. The aim of this model was to measure the connection and evaluate how much hearing forecasts depression. The regression coefficient for hearing was positive and statistically significant ($\beta = 0.0319$, $p < 0.001$), demonstrating that greater difficulties with hearing correlate with higher depression scores. Thus, the statistical significance validates that the relationship identified in the correlation analysis remains when analyzed within a regression model.

Even with this statistical significance, the model's explanatory power was quite restricted. The $\mathbf{R^2}$ value of 0.011 shows that hearing accounts for merely about 1.1% of the variability in depression scores. This indicates that the identified relationship, although steady, lacks practical strength and that a majority of the variation in depression is influenced by other factors.

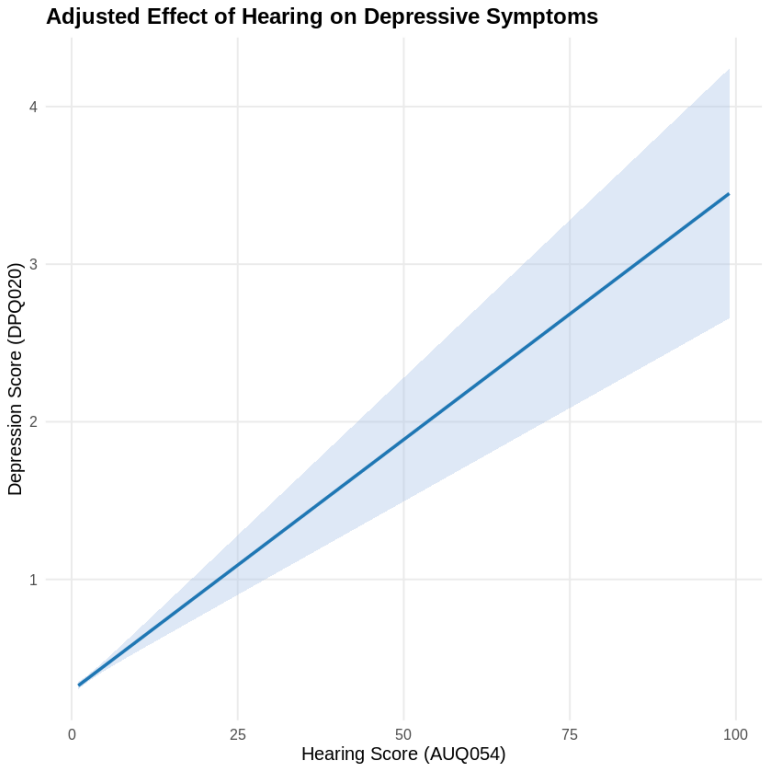


Figure 11: Linear Regression Results

Figure 11 shows linear regression analysis exploring the link between hearing status and depressive symptoms reveals a slight positive correlation. People with worse hearing usually indicated somewhat elevated depression scores, indicated by the upward trend of the regression line. Nevertheless, the data points are significantly scattered around the line, and the confidence interval widens at the extremes, indicating substantial individual variation and diminished certainty in forecasts for exceptional or very poor hearing levels. Although this trend can be measured statistically, the mild slope indicates that hearing by itself accounts for only a limited portion of the variation in depressive symptoms. These findings emphasize the necessity of taking into account other factors, like age, socioeconomic status, and lifestyle, in multivariate models to more effectively forecast depression risk.

Therefore, after making some research, we realized the NHANES data are intrinsically multivariate, and numerous variables of interest may be interconnected, thus, we decided to do the same. A variable like smoking could be linked to both depression and hearing status, possibly skewing the estimated effect seen in basic models. Thus, transitioning to multiple regression enables us to evaluate the individual impact of each predictor with the others held constant, determine if previously noted relationships remain after adjustments, and enhance the total explanatory capability of the models. This phase signifies an organic and essential advancement from initial bivariate examinations to a more thorough multivariable methodology that more accurately captures the intricacies of actual health data, and adjustes for each model presented previously.

3.2 MODEL 1: Multiple Regression based on Age

To address this limitation, Model 1 extends the analysis by incorporating age into a multiple linear regression framework. This adjustment is crucial, as hearing difficulties are strongly related to demographic characteristics, particularly age, and depression patterns differ across population subgroups. After controlling for these variables, hearing status remains a statistically significant predictor of depressive symptoms, and its coefficient slightly increases in magnitude compared to the simple regression. This result demonstrates that the association between hearing difficulties and depression is not solely driven by demographic confounding and supports the hypothesis that hearing problems are independently associated with higher depressive symptom scores.

When age is included in the multiple regression model, the interpretation of the results becomes more nuanced and informative. Hearing status remains a statistically significant predictor of depression ($\beta = 0.117$, $p < 0.001$), and its coefficient slightly increases in magnitude compared to the simple regression. This indicates that, after controlling for age, individuals with hearing difficulties still report higher depressive symptoms, and that this relationship is not explained away by basic demographic differences. The persistence and strengthening of the hearing coefficient supports the central hypothesis of the project: hearing difficulties are independently associated with depressive symptoms in the NHANES population.

Thus, age also emerges as a statistically significant predictor, although its effect size is small and negative (β =-0.0013, $p = 0.033$). This suggests that, holding hearing status, depressive symptom scores slightly decrease with increasing age in this sample. While the magnitude of this effect is minimal, its statistical significance highlights the importance of adjusting for age when studying depression-related outcomes. The presence of an age effect confirms that demographic structure influences depression scores and justifies its inclusion as a control variable in the model.

From a model performance perspective, the multiple regression represents a clear improvement over the simple regression. The R-squared value increases from approximately 0.006 to 0.014, more than doubling the proportion of explained variance. Although the overall explanatory power remains modest, this improvement underscores the multifactorial nature of depression and demonstrates that incorporating demographic variables leads to a more accurate and realistic representation of the data. The statistically significant overall F-statistic, which has a very small p-value, further confirms that the combined predictors provide a better fit than hearing status alone.

The effect plots from Model 1 visually confirm and clarify the results obtained from the multiple regression analysis by illustrating the adjusted relationship between each predictor and depressive symptoms while holding the other variables constant.

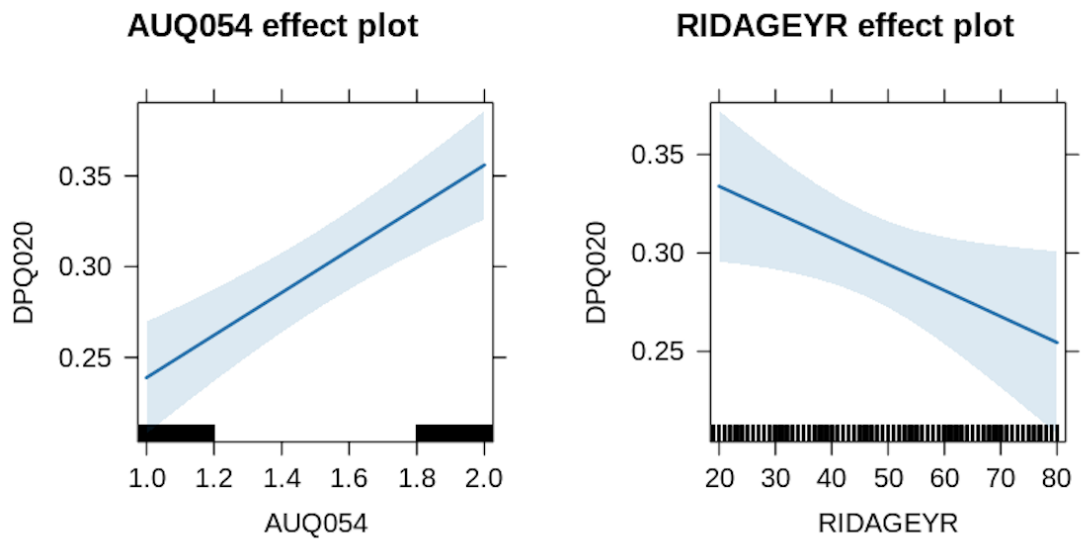


Figure 12: Model 1 Multiple Regression Results

The AUQ054 effect plot (hearing status) (Figure 12) shows a clear positive trend, indicating that individuals reporting hearing difficulties have higher predicted depression scores compared to those without hearing problems. The upward slope reflects the positive regression coefficient observed in the model, and the separation between the confidence bands suggests that this effect is statistically meaningful. Importantly, this plot represents the adjusted effect of hearing, meaning that the increase in depressive symptoms associated with hearing difficulties persists even after controlling for age and sex. This visual evidence reinforces the conclusion that hearing problems are independently associated with higher levels of depressive symptoms.

The RIDAGEYR effect plot (age) (Figure 12) displays a slight downward trend in predicted depression scores as age increases. Although the slope is shallow, it aligns with the statistically significant but small negative coefficient observed in the regression output. The relatively wide confidence interval indicates that the effect size is modest, but the overall direction suggests that, when hearing status and sex are held constant, depressive symptom scores tend to decrease slightly with age in this sample. This plot highlights the role of age as a relevant covariate that influences depression scores and modifies the interpretation of the hearing–depression relationship.

Taken together, these effect plots demonstrate that hearing difficulties are associated with higher depressive symptoms independently of age, while age themselves explain additional variation in depression. The plots therefore complement the numerical regression results by illustrating both the direction and relative magnitude of each effect, strengthening the overall conclusion that depression is influenced by multiple demographic factors.

In conclusion, Model 1 demonstrates that hearing status is significantly associated with depressive symptoms in the NHANES sample, even after controlling for age. The effect plots confirm that individuals reporting hearing difficulties have higher predicted depression scores,

illustrating the independent contribution of hearing to depressive symptoms. Age, while statistically significant, has a very small effect and does not meaningfully modify the relationship between hearing and depression, indicating that its role as a confounding factor is limited in this sample. These results show that hearing difficulties are independently linked to higher depressive symptom scores, while age and sex explain only a modest portion of the variation in depression. Overall, Model 1 highlights that depression is a multifactorial outcome influenced by sensory health and demographic factors, providing a strong foundation for subsequent analyses that incorporate behavioral and socioeconomic variables to further understand contributors to depressive symptoms.

3.3 MODEL 2: Multiple Regression based on Household Income

To address potential confounding, Model 2 extends the analysis by incorporating annual household income into a multiple linear regression framework. This adjustment is important, as hearing difficulties may be related to socioeconomic factors, and depression patterns can differ according to economic status. After controlling for these variables, hearing status remains a statistically significant predictor of depressive symptoms, although its coefficient is slightly smaller in magnitude compared to the simple regression. This result indicates that the association between hearing difficulties and depression is not solely explained by socioeconomic differences, supporting the hypothesis that hearing problems are independently associated with higher depression scores.

Incorporating household income into the multiple regression model yields a more nuanced and informative interpretation. Hearing status continues to be a crucial predictor of depression ($\beta = 0.028$, $p < 0.001$), suggesting that even when controlling for income, those with hearing impairments still experience greater depressive symptoms. Household income also appears as a statistically significant predictor, exhibiting a negative effect (β =-0.023, $p < 0.001$), indicating that, when controlling for hearing status, individuals in higher-income households generally display lower depression scores. The endurance of the hearing effect and the importance of income affirm that each factor independently influences depressive symptoms.

From a model performance perspective, the multiple regression represents an improvement over the simple regression. The R-squared value increases to approximately 0.027, indicating that the proportion of explained variance nearly doubles when household income is included. Although the overall explanatory power remains modest, this improvement underscores the multifactorial nature of depression and shows that considering socioeconomic variables leads to a more realistic representation of the data. The global F-statistic, highly significant with a very small p-value, further confirms that the combined predictors provide a better fit than hearing status alone.

Effect plots from Model 2 visually confirm and clarify the multiple regression results by illustrating the adjusted relationship between each predictor and depressive symptoms.

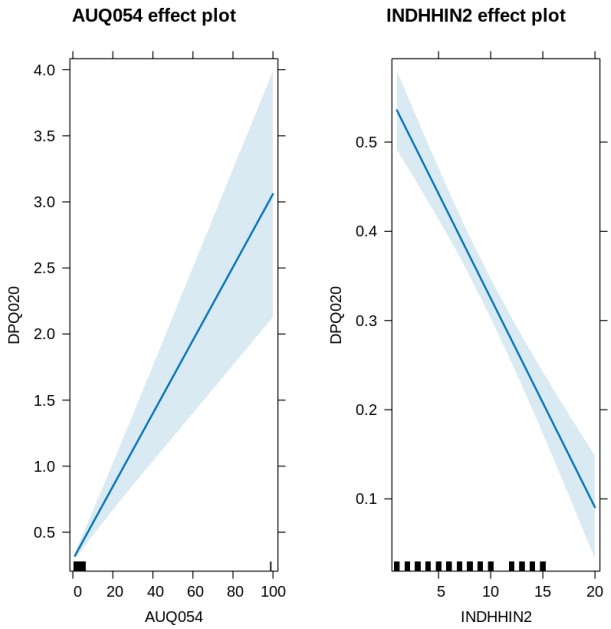


Figure 13: Model 2 Multiple Regression Results

Again, the AUQ054 effect plot (Figure 13) illustrates a strong positive relationship between hearing difficulties (AUQ054) and depressive symptoms (DPQ020). As AUQ054 increases from 0 to 100, the predicted depression score rises substantially, from approximately 0.4 to over 3.0. The solid blue line represents the predicted change, while the light blue shaded area indicates the 95% confidence interval. Notably, the confidence interval widens at higher values of AUQ054, indicating greater uncertainty in predictions where fewer data points exist, as shown by the black rug plot at the bottom. Overall, this plot confirms that hearing difficulties are a strong predictor of depressive symptoms, with higher levels of AUQ054 associated with higher predicted DPQ020 scores.

On the other side, the INDHHIN2 effect plot (Figure 13) shows a negative relationship between household income (INDHHIN2) and depressive symptoms (DPQ020). As income increases from 0 to 20, the predicted depression score decreases from roughly 0.53 to about 0.1. The light blue shaded area remains relatively consistent across the range, suggesting stable predictions. The rug plot at the bottom indicates a fairly even distribution of data points, with a slight gap between values 10 and 12. Compared to AUQ054, the total change in predicted depression is much smaller, indicating that household income has a weaker overall impact on depressive symptoms. Nevertheless, the negative trend confirms that higher income is associated with lower predicted DPQ020 scores.

Together, these plots highlight the differential effects of the two predictors. Hearing difficulties (AUQ054) have a strong positive association with depressive symptoms, while household income (INDHHIN2) has a smaller, negative association. The widening confidence interval in AUQ054 at higher values emphasizes the importance of data density in prediction certainty, whereas INDHHIN2 predictions are more stable across its range. These visualizations reinforce the findings of the regression model, demonstrating that depression is influenced by multiple factors, including sensory health and socioeconomic status.

In conclusion, Model 2 shows that hearing status remains significantly associated with depressive symptoms in the NHANES sample, even after controlling for household income. The effect plots confirm that individuals with hearing difficulties have higher predicted depression scores, illustrating the independent contribution of hearing to depressive symptoms. Household income is also a significant predictor with a negative effect, indicating that higher income is associated with lower depressive symptoms.

3.4 MODEL 3: Multiple Regression based on Smoking Habits

To further explore behavioral factors that may influence depressive symptoms, Model 3 extends the analysis by incorporating smoking habits into a multiple linear regression framework. Smoking is a relevant variable to consider, as it has been consistently associated with both mental health outcomes and health-related behaviors, and may confound or modify the relationship between hearing difficulties and depression. This model therefore examines whether the association between hearing status and depressive symptoms persists after accounting for smoking behavior.

In the initial regression including only hearing status, AUQ054 remains a statistically significant predictor of depressive symptoms ($\beta = 0.107$, $p = 0.008$), indicating that individuals reporting greater hearing difficulties tend to report higher depression scores. However, the very low R-squared value (approximately 0.005) suggests that hearing status alone explains only a small proportion of the variability in depressive symptoms, reinforcing the need to consider additional behavioral covariates such as smoking.

When smoking habits (SMD030) are added to the model, hearing status continues to show a statistically significant positive association with depressive symptoms ($\beta = 0.025$, $p < 0.001$). Although the magnitude of the hearing coefficient decreases compared to the simple regression, its persistence indicates that the relationship between hearing difficulties and depression is not fully explained by smoking behavior. This attenuation suggests some shared variability between hearing status and smoking, but confirms that hearing difficulties remain independently associated with higher depressive symptom scores.

Smoking habits themselves do not emerge as a statistically significant predictor in the multiple regression model ($\beta = -0.0004$, $p = 0.18$). While the estimated coefficient is negative, indicating a slight decrease in predicted depression scores with increasing smoking exposure, this effect is small and not statistically distinguishable from zero. This result suggests that, within this sample and model specification, smoking does not have a strong independent association with depressive symptoms once hearing status is taken into account.

From a model performance perspective, the inclusion of smoking leads to a modest improvement in explanatory power. The R-squared value increases to approximately 0.008, indicating that the combined predictors explain slightly more variance in depressive symptoms than hearing status alone. Although the overall explanatory power remains low, the statistically significant F-statistic confirms that the model provides a better fit than a simple regression. This modest increase further highlights the multifactorial nature of depression and the difficulty of capturing

its complexity using a limited set of predictors.

The effect plots from Model 3 visually illustrate the adjusted relationships between hearing status, smoking habits, and depressive symptoms while holding the other variable constant.

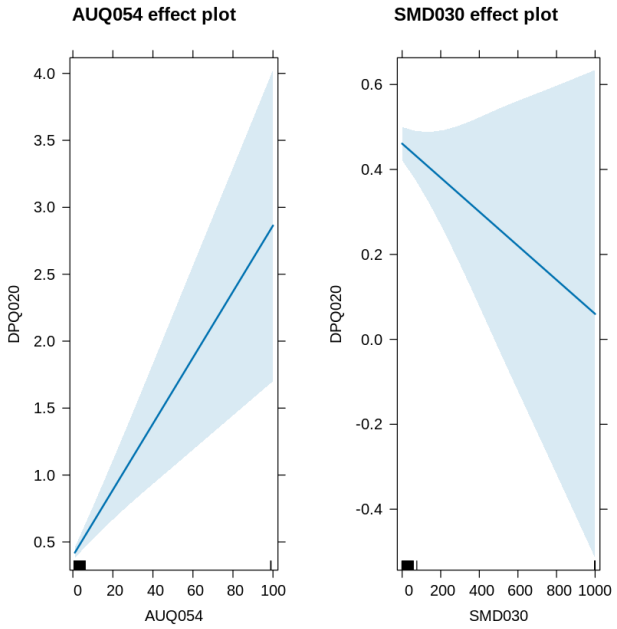


Figure 14: Model 3 Multiple Regression Results

In this case, the SMD030 effect (Figure 14) displays a shallow negative slope, indicating a weak inverse relationship between smoking exposure and predicted depressive symptoms. The confidence interval is relatively wide and overlaps substantially across the range of SMD030 values, visually reinforcing the lack of statistical significance observed in the regression results. This suggests that smoking habits, while potentially related to depression through other pathways, do not show a strong independent effect in this model once hearing status is accounted for.

Taken together, these effect plots confirm that hearing difficulties remain the dominant predictor of depressive symptoms in Model 3, while smoking habits contribute little additional explanatory power. The visualizations complement the numerical regression results by clearly illustrating both the direction and relative magnitude of each effect.

In conclusion, Model 3 demonstrates that hearing status is significantly associated with depressive symptoms in the NHANES sample even after controlling for smoking habits. The effect plots confirm that individuals with greater hearing difficulties have higher predicted depression scores, highlighting the independent contribution of hearing to mental health outcomes. Smoking habits do not show a statistically significant independent association with depressive symptoms in this model, suggesting that their role may be secondary or mediated through other factors not captured here.

Taken together, the results from Models 1–3 indicate that the relationship between hearing

difficulties and depressive symptoms is robust across demographic, socioeconomic, and behavioral contexts. Although age, household income, and smoking habits are relevant variables and explain a small portion of the variability in depression scores, none of them eliminate or substantially attenuate the association between hearing status and depressive symptoms. This suggests that hearing difficulties are independently associated with higher levels of depression and that the observed relationship is not merely a byproduct of demographic structure, socioeconomic conditions, or health-related behaviors. While these additional variables provide important contextual information, hearing status remains the most consistent predictor across all models.

4 Depression difference among Hearing Condition Groups

Descriptive Statistics Descriptive statistics were calculated to examine depressive symptoms across levels of self-reported hearing condition. Depression symptoms were measured using two items from the Patient Health Questionnaire: lack of interest or pleasure in doing things (DPQ010) and feeling down, depressed, or hopeless (DPQ020). Higher scores indicate more frequent depressive symptoms.

Table 1 presents the mean depression scores for each hearing condition group.

Table 1: Mean Depression Scores by Hearing Condition

Hearing Condition	Mean DPQ010	Mean DPQ020	N
Excellent	0.27	0.24	1882
Good	0.39	0.34	2003
Little trouble	0.56	0.50	730
Moderate trouble	0.54	0.61	321
A lot of trouble	0.66	0.53	128
Deaf	0.64	0.57	14

As shown in Table 1, mean depression scores increased as hearing condition worsened. Participants reporting excellent or good hearing had the lowest levels of depressive symptoms, whereas those reporting hearing difficulties exhibited substantially higher mean scores on both depression measures.

4.1 One-Way ANOVA

In order to check whether depression differed significantly across hearing condition groups, an ANOVA test was done, in order to determine if an independent variable affects a dependent variable across different categories. To do so, DPQ010 and DPQ020 have been used.

Anova Results For DPQ010, the one-way ANOVA revealed a significant effect of hearing condition on lack of interest or pleasure in doing things, $F(5, 5072) = 23.31$, $p < .001$. Similarly, the ANOVA for DPQ020 indicated a significant effect of hearing condition on feelings of depression or hopelessness, $F(5, 5072) = 25.93$, $p < .001$.

These findings indicate that **depressive symptoms vary significantly across different levels of self-reported hearing conditions**.

4.2 Post-hoc Analyses

Post-hoc pairwise comparisons were conducted using Tukey's test to identify specific group differences.

For DPQ020, **participants with excellent hearing reported significantly lower depression scores compared to those with little trouble, moderate trouble, and a lot of trouble hearing (all** $p < .001$**)**. Significant differences were also observed between the good hearing group and multiple hearing-impaired groups. Comparisons involving the deaf group were not statistically significant, due to the small sample size of this group.

For DPQ010, significant differences were primarily observed between participants with good hearing and those reporting moderate or severe hearing trouble, suggesting that the presence of hearing impairment is associated with increased depression.

4.3 t-Tests

To further examine the relationship between hearing impairment and depressive symptoms, hearing condition was represented into good hearing (excellent or good) and hearing trouble (any reported hearing difficulty or deafness).

For DPQ020, **individuals with hearing trouble reported significantly higher depression scores** ($M = 0.53$) **than those with good hearing** ($M = 0.29$), $t(1603.3) = -8.65$, $p < .001$, 95% CI $[-0.29, -0.18]$.

Similarly, DPQ010 scores were **significantly higher among participants with hearing trouble (**$M = 0.57$**) compared to those with good hearing** ($M = 0.33$), $t(1641.9) = -8.16$, $p < .001$, 95% CI $[-0.29, -0.18]$.

These results confirm that **individuals with worse hearing conditions have significantly greater depressive symptoms than those without hearing difficulties**.

4.4 Visualization of Depression Scores

Figure 15 presents boxplots of DPQ020 scores across hearing condition groups. The figure illustrates a clear upward trend in depression severity as hearing condition worsens, supporting the results of the descriptive and inferential analyses.

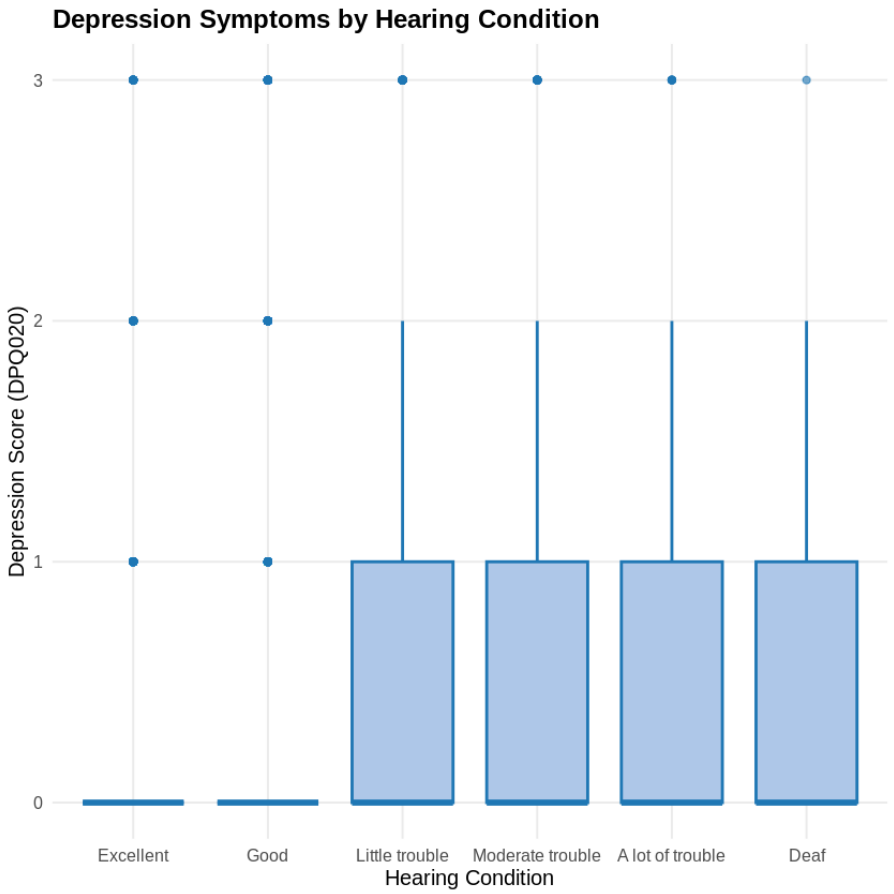


Figure 15: Distribution of DPQ020 scores by hearing condition

The analyses conducted in this section provide strong evidence that depression symptoms depend across self-reported hearing condition groups. This study proved that **mean depression scores increased progressively as hearing condition worsened**.

Participants reporting excellent or good hearing exhibited the lowest levels of depressive symptoms, while those reporting degrees of hearing difficulty showed higher scores on both depression indicators (DPQ010 and DPQ020).

One-way ANOVA results demonstrated that hearing condition has a statistically significant effect on depressive symptoms for both measures, indicating that the differences across groups are unlikely to be due to random variation.

Post-hoc comparisons showed that individuals with hearing difficulties—particularly those reporting little to moderate trouble—had significantly higher depression scores compared to participants with excellent or good hearing.

t-Test analyses reinforced these results by simplifying hearing status into two categories: good hearing and hearing trouble. Individuals with hearing trouble reported significantly higher levels of depressive symptoms on both depression measures, with large and highly significant mean differences.

Although comparisons involving the deaf group were limited by small sample size, the overall pattern remained consistent with the findings.

5 Observed problems

Through the analysis of this project, several limitations and methodological challenges were observed. Considering them carefully for interpreting correctly the findings.

Magnitude of the observed associations was relatively small Correlation and regression analyses showed low coefficients, showing that hearing condition alone explains only a small proportion of the variability in depression scores. In order to solve this issue, multivariate regression and correlation were used including other variables, such as socioeconomic or habits.

The use of self-reported measures may include different challenges for both hearing condition and depressive symptoms. Participant's answers may not be accurate or supported by a professional in the field, leading into bias. While NHANES is a high-quality dataset, self-reported variables inherently reduce measurement precision compared to clinical diagnoses or objective audiometric thresholds.

Imbalance in group sizes across hearing condition categories particularly for severe hearing loss and deafness, since these groups contained fewer observations, which reduced statistical accuracy and limited comparisons. As a result, some differences may not have reached statistical significance despite showing consistent trends.

Age xacted as a major confounding factor Hearing loss is strongly associated with aging, while depression exhibits different patterns across the age. The stratified age analysis revealed that the relationship between hearing impairment and depression was stronger among younger adults and weaker among older individuals. In order to solve this issue, during the analyses different age groups were divided.

To sum up, although the results consistently support the hypothesis that worse hearing conditions are associated with increased depressive symptoms, these findings should be interpreted with caution. Different factors must be taken into account, such as the magnitude of the samples, imbalances in group sizes and age factors.

6 Conclusion

This project examined the relationship between hearing condition and depressive symptoms using data from the National Health and Nutrition Examination Survey (NHANES). The analysis incorporated demographic, socioeconomic, and behavioral variables in order to assess how these factors influence the association between hearing difficulties and depression. By including age, household income, and smoking habits, the study evaluated whether the observed relationship between hearing condition and depressive symptoms persisted after accounting for potential confounding factors.

Across all analytical approaches, **the central hypothesis that hearing difficulties are associated with higher levels of depressive symptoms was consistently supported**. Exploratory analyses, group comparisons, and regression models all revealed a **positive and systematic trend in which worsening hearing condition corresponded to higher average depression scores**. These findings indicate that individuals with poorer hearing are more likely to experience depressive symptoms.

Multivariate regression analyses demonstrated that **hearing status remains a statistically significant predictor of depressive symptoms** even after controlling for age, income, and smoking behavior. Age and household income emerged as relevant covariates: **older age was associated with slightly lower depression scores**, and **higher income showed a protective effect against depressive symptoms**. In contrast, smoking habits did not display a statistically significant independent association once hearing status was taken into account. Importantly, **none of these additional variables eliminated or substantially weakened the association between hearing difficulties and depression**, indicating that hearing status is independently linked to depressive symptoms.

Further support for this relationship was provided by the group-based analyses presented in Section 4. Using **one-way ANOVA, post-hoc comparisons, and t-tests**, the study identified significant differences in depressive symptoms across hearing condition categories. These analyses consistently showed that **participants with hearing trouble reported significantly higher levels of depressive symptoms than those with good or excellent hearing**. The convergence of results across multiple statistical tests strengthens the robustness of the findings.

Despite the consistency of the results, the overall explanatory power of the statistical models remained limited, reflecting the **multifactorial nature of depression**. Several limitations were identified, including **imbalanced group sizes for severe hearing loss categories** and the **cross-sectional design of the NHANES dataset**. Additionally, stratified analyses suggested that **the relationship between hearing difficulties and depression may vary across age groups**, indicating the need for further research to better understand how this association evolves over the life course.

In conclusion, this study provides **statistical evidence that hearing difficulties are meaningfully associated with higher depressive symptoms in the U.S. population**. Although hearing loss alone does not explain most of the variability in depression, it emerges as a **stable and independent risk factor**. These findings highlight the importance of consider-

ing sensory health within mental health research and suggest that **early identification and support for individuals with hearing difficulties may play a role in mitigating depressive symptoms**. Future research incorporating longitudinal data and objective hearing measures would help clarify the mechanisms underlying this relationship.

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